MODULE - 3

ALTERNATING CURRENT

Generation of Alternating voltage and current

Con. a coil of N turns rotating in anticlockwise direction with an angular velocity of ω radians / sec in a uniform mag field as shown

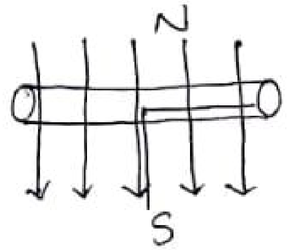


fig (1)

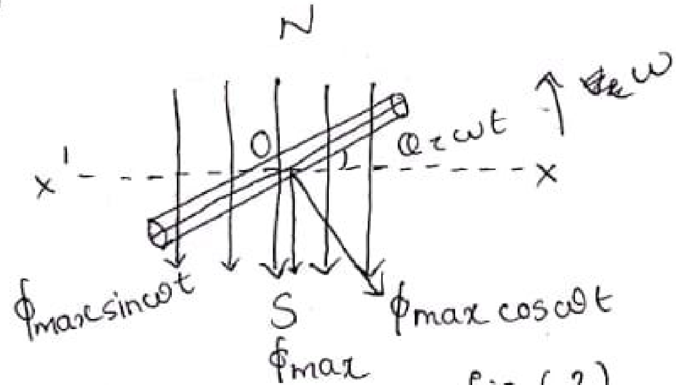


fig (2)

let the time be measured from the instant the plane of coil coincide with ox axis.

In fig (1) let the flux linking with the coil has a max value of ϕ_{max}. let the coil turn through an angle $\theta = \omega t$ in anticlockwise direction in t Seconds and as shown in fig (2). In this position the max flux ϕ_m acting vertically downward can be resolved into two $\perp r$ components $\phi_m \sin \omega t$ and $\phi_m \cos \omega t$

Component $\phi_m \sin \omega t$ $\parallel$el to the plane of coil, this component induces no emf in the coil and component $\phi_m \cos \omega t$ $\perp r$ to the plane of coil induces emf in the coil

Flux linkage of the coil at considered instant

i.e ϕ = no of turns X flux linking component

$= 5\sin\left(2\times 3.14\times \cancel{50}\times \frac{1}{\cancel{300}_{6}}\right) A$

$=$

Q. An alternating current is given by $I = 50\sin(314t)$ find max value of freq and timeperiod

ans $2\pi\nu = 314$ $\nu = \frac{3.14}{2\times\pi} = \frac{\cancel{314}\,100}{2\times\cancel{3.14}} = 50\,Hz$

$t = \frac{1}{f} = \frac{1}{50} = 0.02s$

Q. A sinusoidal Alternating current has a peak value of 20A at 50Hz how long will it take current to reach 10A starting from zero.

$I = I_m \sin(\omega t)$ $I = I_m\sin(2\pi\nu t)$

$\cancel{10} = \cancel{20}\sin(2\times 180\times 50\times t)$

$\sin^{-1}\left(\frac{1}{2}\right) = 2\times 180\times 50\times t$ $\frac{\cancel{180}}{6} = 2\times\cancel{180}\times 50\times t$

$t = \frac{1}{6\times 2\times 50} = \frac{1}{600}$

Average Value

The avg value of an alternating quantity is defined as steady state current which transfer across any circuit the same amount of charge as it transfered by an alternating current during the same time.

The avg value of a waveform is known as its DC value

Average value is also defined as total area measured over half cycle divided by its base for a periodic waveform gives average value.

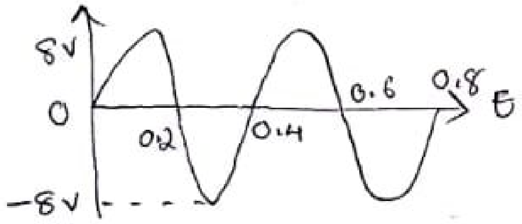
Period

The time taken for one cycle is known as time period. Denoted as 'T' $T = \frac{1}{f}$ $f \rightarrow$ frequency

Frequency

No of cycles completed in one second. It is expressed in hertz or cycles / second.

Q. Con a sinusoidal waveform shown below



i. find peak value ii. Instantaneous value at 0.3 sec and 0.6 sec iii. find peak to peak value iv. find period of wave form v. How many cycles are shown in figure vi. find freq of wave form

ans i. peak value = 8V

ii. Instantaneous at 0.3 = -8

0.6 = 0

iii. 16 iv. 0.4 sec v. 2 cycles

vi. $\frac{1}{0.4} = 2.5\ Hz$

Q. A sinusoidal Aalternating current having a freq of 50 Hz has a peak value of 5 A what is the value of current after 1/300 sec from zero.

ans. A = 5 A freq = 50 Hz

$I = I_m \sin \omega t = I_m \sin(2\pi \nu t)$

$$V_{rms} = \sqrt{\frac{V_m^2}{\pi}\int_0^{\pi}\frac{(1-\cos 2\theta)}{2}} = \sqrt{\frac{V_m^2\,\frac{1}{2}\int_0^{\pi}1\,d\theta - \frac{1}{2}\int_0^{\pi}\cos 2\theta\,d\theta}{\pi}}$$

$$= \sqrt{\frac{V_m^2\left[\frac{1}{2}[\theta]_0^{\pi} - \frac{1}{2}\left[\frac{\sin 2\theta}{2}\right]_0^{\pi}\right]}{\pi}} = \sqrt{\frac{V_m^2\left[\frac{1}{2}(\pi-0) - \frac{1}{4}(\sin 2\pi - \sin 0)\right]}{\pi}}$$

$$= \sqrt{\frac{V_m^2\left[\frac{1}{2}\pi - \frac{1}{4}(0-0)\right]}{\pi}} = \sqrt{\frac{V_m^2\,\pi/2}{\pi}} = V_m/\sqrt{2}\ //$$

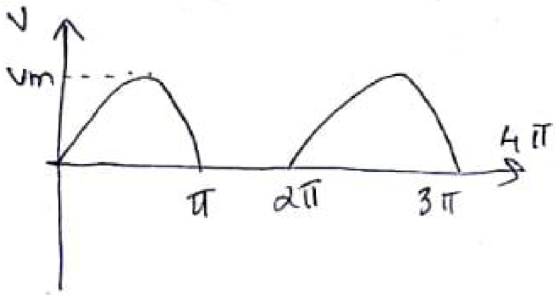
~~form~~ peak factor $= \dfrac{\text{peak } \cancel{\text{avg}} \text{ value}}{\text{rms value}} = \dfrac{V_m}{V_m/\sqrt{2}} = \sqrt{2}\ //$

$= 1.412\ //$

form factor $= \dfrac{V_m}{\sqrt{2}} \times \dfrac{\pi}{2V_m} = \dfrac{\pi}{2\sqrt{2}} = \dfrac{\sqrt{2}\pi}{4}\ //$

$= 1.11\ //$

Q. find rms, avg and form factor of a half wave rectifier



ans avg value = $\dfrac{\text{area of cycle}}{\text{base}} = \int_0^{2\pi}\dfrac{V_m\sin\theta}{2\pi}$

$V_{avg} = \dfrac{\cancel{2}V_m}{\pi}\ //$

$V_{rms} = V_m/\sqrt{2}\ //$

form factor $= \dfrac{V_m}{\sqrt{2}} \times \dfrac{\pi}{2V_m} = \dfrac{\pi}{2\sqrt{2}} = \dfrac{\sqrt{2}\pi}{4}\ //$

ψ Φ = $N \phi_m \cos\omega t$

According to Faraday's law of EMI. Emf at considered instant $e = -N\frac{d\phi}{dt} - \frac{d\psi}{dt}$

$$e = -N \frac{d}{dt} \phi_m \cos\omega t$$

$$= -N\phi_m - \sin\omega t \, \omega \quad = N\phi_m \omega \sin\omega t$$

The value of e is maximum when $\sin\omega t = 1$

i.e $\omega t = 90°$ from referance axis

Max value of $e = N\phi_m \omega$

$$\boxed{e = e_m \sin\omega t}$$

Instantaneous Value

The value of voltage or current at a particular instant. Eg:- $e = e_m \sin\omega t$ $\quad I = I_m \sin\omega t$

Amplitude

It is the magnitude of maximum +ve or -ve value of alternating quantity it is also known as peak value.

Peak to peak value

Full voltage between +Ve and -ve value peak of wave form

Periodic Waveform

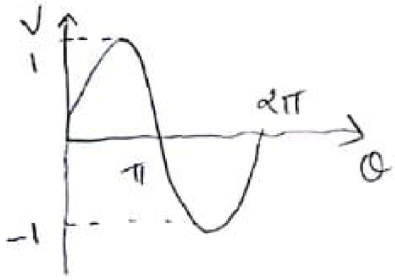
A waveform that continuously repeat itself after same interval.

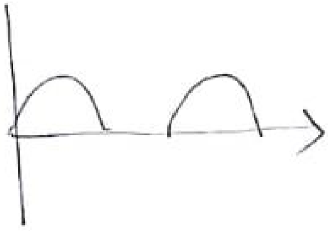
Cycle.

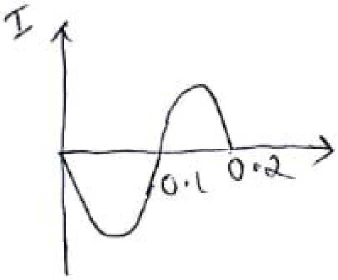
One complete set of +Ve and -ve value of an alternating quantity is known as cycle.

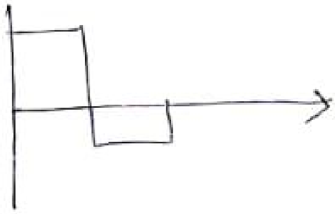
Symmetrical

unsymmetrical



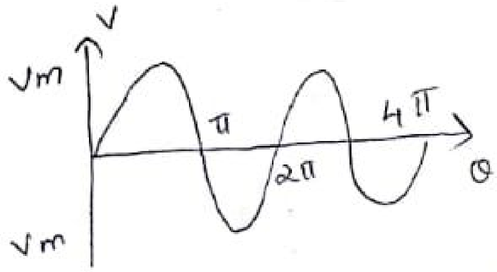






Q. On a sine wave find avg value, rms, peak value
form factor

ans



avg value $= \frac{\text{area of half cycle}}{\text{base}} = \frac{\int_0^\pi V_m \sin\theta}{\pi}$

$$V_{avg} = \frac{V_m}{\pi}\int_0^\pi \sin\theta = \frac{V_m}{\pi}\left[-\cos\theta\right]_0^\pi$$

$$V_{av} = -\frac{V_m}{\pi}\left[\cos\pi - \cos 0\right] = -\frac{V_m}{\pi}\left[-1-1\right] = -\frac{V_m}{\pi}\times 2$$

$$\boxed{V_{avg} = \frac{2V_m}{\pi}}$$

RMS value

$$V_{rms} = \sqrt{\frac{\int_0^\pi (V_m \sin\theta)^2}{\pi}}$$

$$V_{rms} = \sqrt{\frac{\int_0^\pi V_m^2 \sin^2\theta}{\pi}} = \sqrt{\frac{V_m^2}{\pi}\int_0^\pi \sin^2\theta}$$

RMS Value

RMS value of an alternating quantity may be defined as value of DC current when flows through a given resistance produce the same amount of heat as that produced by alternating current passing through same resistance in same time.

Heat is ∝ to square of current

∴ AC value is squared first then avg or mean value is found out finally root is taken to give effective value.

Form Factor

It is defined as the ratio of rms value to average value

Peak factor

It is defined as ratio of its max value to rms value

for a symmetrical or periodic waveform. the area above horizontal axis is +ve and area below horizontal axis is -ve the result is that avg or mean value of a symmetrical alternating quantity is zero because it cancel each other.

NOTE

For a symmetrical wave form the avg value ~~of~~ over complete cycle is zero ∴ the average value is defined for half cycle only.

For unsymmetrical wave form the full cycle should be considered for finding avg value or rms value

$$= \frac{50[\sqrt{2}+1]}{\sqrt{2}\,\pi} = \frac{50\sqrt{2}+50}{\sqrt{2}\,\pi} = \frac{100+50\sqrt{2}}{2\pi} = 27.2$$

$$V_{rms} = \sqrt{\frac{\int_{\pi/4}^{\pi} 100\sin^2\theta\, d\theta}{2\pi}} = \sqrt{\frac{100^2\int_{\pi/4}^{\pi}\frac{1-\cos 2\theta}{2}\,d\theta}{2\pi}}$$

$\pi - \frac{\pi}{4}$

$\frac{4\pi-\pi}{4}$

$50/4$

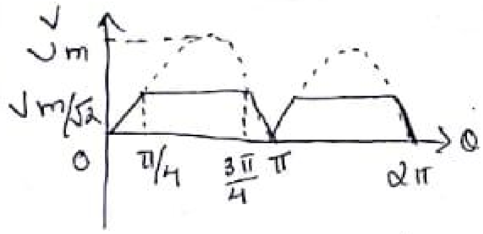
$$= \sqrt{\frac{100^2\left(\frac{1}{2}[\theta]_{\pi/4}^{\pi} - \frac{1}{2}\left[\frac{\sin 2\theta}{2}\right]_{\pi/4}^{\pi}\right)}{2\pi}}$$

$$= 100\sqrt{\frac{\frac{3\pi}{8} + \frac{1}{2}\cdot\frac{1}{2}}{2\pi}} = 100\sqrt{\frac{\frac{3\pi}{8}+\frac{1}{4}}{2\pi}}$$

$$= \cancel{100\sqrt{\frac{3\pi-1}{4\times 2\pi}}} = \cancel{100\sqrt{\frac{3\pi-1}{8\pi}}} = \cancel{57.84}$$

$$= \cancel{100}\ 47.67$$

Q. A full wave rectified sine wave is cliped at 0.707 of its max value as shown in fig find avg value.



$$= \frac{\int_0^{\pi/4} V_m\sin\theta + \int_{\pi/4}^{3\pi/4} V_m/\sqrt{2}\, d\theta + \int_{3\pi/4}^{\pi} V_m\sin\theta}{\pi}$$

$$= \frac{V_m[-\cos\theta]_0^{\pi/4} + \frac{V_m}{\sqrt{2}}\left[3\pi/4 - \pi/4\right] + V_m[-\cos\theta]_{3\pi/4}^{\pi}}{\pi}$$

$$= \frac{V_m \times -1/\sqrt{2} + \frac{V_m}{\sqrt{2}}\frac{2\pi}{4} + V_m[+1+1/\sqrt{2}]}{\pi}$$

$$\text{Avg value} = \frac{\int_0^{2\pi} V\,d\theta}{2\pi} = \frac{\int_0^{2\pi} V_m \sin\theta + \int_\pi^{2\pi} 0\,d\theta}{2\pi}$$

$$= \frac{V_m[-\cos\theta]_0^\pi + 0}{2\pi} = \frac{V_m[-\cos\pi + \cos 0]}{2\pi}$$

$$= \frac{V_m[+1+1]}{2\pi} = \frac{2V_m}{2\pi} = V_m/\pi$$

$$\text{RMS Value} = \sqrt{\frac{\int_0^\pi (V_m \sin\theta)^2\,d\theta}{2\pi}} = \sqrt{\frac{V_m^2 \int_0^\pi \left(\frac{1-\cos 2\theta}{2}\right) d\theta}{2\pi}}$$

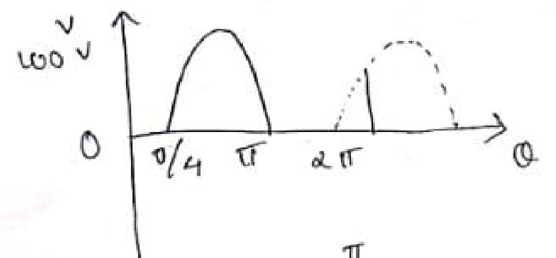
$$= \sqrt{\frac{V_m^2 \left[\frac{1}{2}[\theta]_0^\pi - \frac{1}{2}\frac{\sin 2\theta}{2}\right]_0^\pi}{2\pi}}$$

$$= \sqrt{\frac{V_m^2 \left[\frac{1}{2}(\pi) - \frac{1}{4}(\sin 2\pi)\right]}{2\pi}} = \sqrt{\frac{V_m^2 \pi/2}{2\pi}} = \sqrt{\frac{V_m^2 \cancel{\pi}}{4\cancel{\pi}}}$$

$$= \frac{V_m}{2}$$

form factor $V_m/2 \times \frac{\pi}{V_m} = \pi/2$

Q. find avg and rms value of given waveform



$$\text{avg value} = \frac{\int_{\pi/4}^{\pi} 100 \sin\theta\,d\theta}{2\pi}$$

$$= \frac{100\,[-\cos\theta]_{\pi/4}^{\pi}}{2\pi} = \frac{100\,[-\cos\pi + \cos\pi/4]}{2\pi}$$

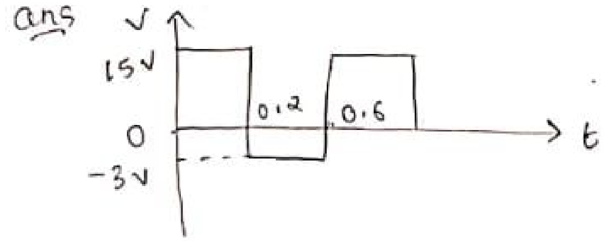
$$= \frac{100 \times [+1 + 1/\sqrt{2}]}{2\pi} = \frac{100\left[\frac{\sqrt{2}+1}{\sqrt{2}}\right]}{2\pi} = \frac{\overset{50}{\cancel{100}}[\sqrt{2}+1]}{\sqrt{2} \times \cancel{2}\,\pi}$$

$$= \frac{-\frac{V_m}{\sqrt{2}} + \frac{\cancel{4} 2 V_m \pi}{2 \cancel{4} \sqrt{2}} + V_m \frac{\sqrt{2}+1}{\sqrt{2}}}{\pi} = V_m \left[\frac{\frac{-1}{\sqrt{2}} + \frac{\pi}{2\sqrt{2}} + \frac{\sqrt{2}+1}{\sqrt{2}}}{\pi} \right]$$

$$= V_m \times 0.518$$

Q. Find avg value of given waveform

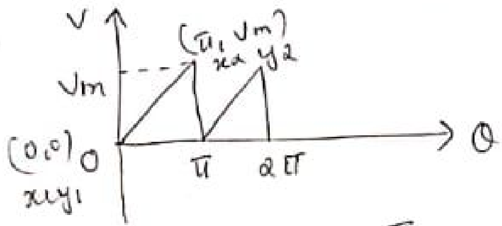
ans



$$\text{avg value} = \frac{15 \times 0.2 + -3 \times 0.4}{0.6}$$

$$= \frac{3.0 - 1.2}{0.6} = \frac{1.8}{0.6} = 3V$$

Q. Calculate avg value and rms value of saw tooth wave form having max value V_m



$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

$$\frac{V - 0}{V_m - 0} = \frac{\theta - 0}{\pi - 0} = \frac{V}{V_m} = \frac{\theta}{\pi}$$

$$V = \frac{V_m \theta}{\pi}$$

$$\text{avg value} = \frac{\int_0^{\pi} \frac{V_m \theta}{\pi} \, d\theta}{\pi}$$

$$= \frac{\frac{V_m}{\pi} \int_0^{\pi} \theta \, d\theta}{\pi} = \frac{\frac{V_m}{\pi} [\theta]_0^{\pi}}{\pi} = \frac{\frac{V_m}{\cancel{\pi}} \times \cancel{\pi}}{\pi} = V_m/\pi$$

$$= \frac{\frac{1}{2} \text{length} \times \text{breadth}}{\text{base}} = \frac{1}{2} \times \frac{V_m \times \cancel{\pi}}{\cancel{\pi}} = V_m/2$$

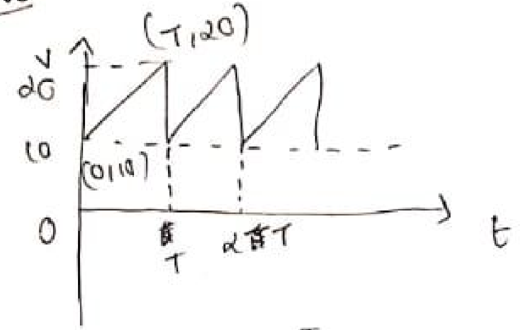
$$\text{rms} = \sqrt{\frac{\int_0^{\pi} \left(\frac{V_m \theta}{\pi}\right)^2 d\theta}{\pi}} = \sqrt{\frac{\int_0^{\pi} \frac{V_m^2 \theta^2}{\pi^2} d\theta}{\pi}}$$

$= \frac{V_m}{\pi}\sqrt{\frac{\int_0^{\pi} \theta^2 d\theta}{\pi}} = \frac{V_m}{\pi}\sqrt{\frac{\cancel{\pi^2}}{\cancel{\pi}}} = \frac{V_m}{\pi} \times \frac{\sqrt{\pi}}{3}$

$= \frac{V_m}{\sqrt{\pi}}$

$\frac{V_m}{\pi}\sqrt{\frac{\left[\frac{\theta^3}{3}\right]_0^{\pi}}{\pi}} = \frac{V_m}{\pi}\sqrt{\frac{\pi^{\cancel{3}2}}{3\cancel{\pi}}} = \frac{V_m}{\cancel{\pi}} \times \frac{\cancel{\pi}}{3} = V_m/3$

H·W



$\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$

$= \frac{y-10}{20-10} = \frac{t-0}{T-0} = \frac{y-10}{10} = \frac{t}{T}$

Avg value $= \frac{\int_0^T y\,dt}{T}$

$y - 10 = \frac{10t}{T} \quad y = \frac{10t}{T} + 10$

$= \frac{\int_0^T \left(\frac{10t}{T} + 10\right) dt}{T} = \frac{\int_0^T \frac{10t\,dt}{T} + \int_0^T 10\,dt}{T}$

$= \frac{\frac{10}{T}\left[\frac{t^2}{2}\right]_0^T + 10\left[t\right]_0^T}{T} = \frac{\frac{10}{\cancel{T}} \times \frac{T^{\cancel{2}}}{2} + 10 \times T}{T}$

$= \frac{\frac{10T}{2} + 10T}{T} = \frac{10}{2} + 10 = 15$

Rms value $= \sqrt{\frac{\int_0^T \left(\frac{10t}{T} + 10\right)^2 dt}{T}} = \sqrt{\frac{\int_0^T \left(\frac{100t^2}{T^2} + 100 + 2 \times \frac{10t}{T} \times 10\right) dt}{T}}$

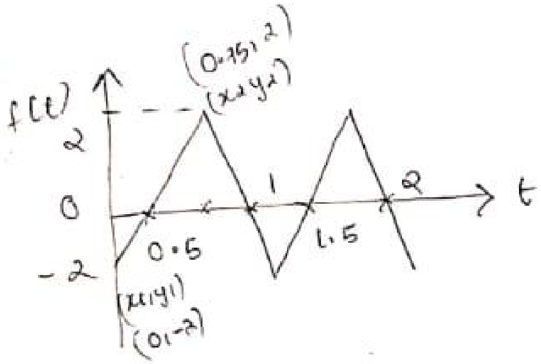
$= \sqrt{\frac{\int_0^T \left(\frac{100t^2}{T^2} + 100 + \frac{200t}{T}\right) dt}{T}} = \sqrt{\frac{\frac{100}{T^2}\left[\frac{t^3}{3}\right]_0^T + 100\left[t\right]_0^T + \frac{200}{T}\left[\frac{t^2}{2}\right]_0^T}{T}}$

$$= \frac{100}{T^2}\frac{T^3}{3} + 100 \times T + \frac{200}{T} \times \frac{T}{2}$$

$$\sqrt{\frac{\frac{100T}{3} + 100T + 200T}{T}} = \sqrt{\frac{100}{3} + 100 + 200}$$

$$= \sqrt{\frac{100}{3} + 300} = 15.27$$

H.W



find avg and rms value

avg value = -1

rms = 1.148

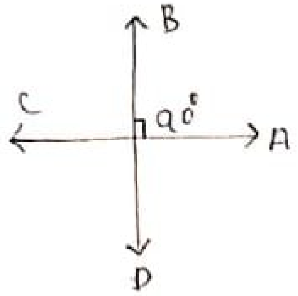
Phasor Representation

An alternating qty can be represented by a rotating phasor. A phasor is a vector rotating with constant angular velocity ω. The phase representations are of 4 type

i. Rectangular form
ii. Polar form
iii. Trignometric form
iv. Exponential form

i Operator

The letter i is used to express the operation of anticlockwise rotation of a vector through 90°.
If this operation is done twice on a vector the vector gets rotated anticlockwise through 180° means the vector reversed its sign.



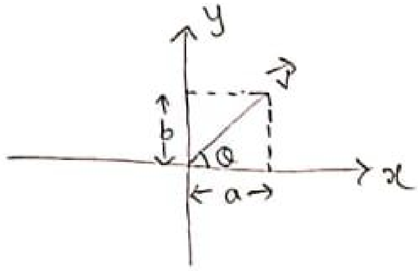
$i = 1\angle 90^\circ$

$\vec{OB} = i\,\vec{OA}$

$\vec{OC} = -\vec{OA}$ $\qquad$ $\vec{OC} = \vec{OA} \times i \times i$

$= \vec{OA}\, i^2 = -\vec{OA}$

ii. Rectangular form



$\boxed{\vec{V} = a + ib}$ $\qquad$ $\theta = \tan^{-1} b/a$

$\vec{V} = \sqrt{a^2+b^2}$

iii. Polar form

$\boxed{\vec{V} = |\vec{V}| \angle \theta}$ $\qquad$ $\theta = \tan^{-1} b/a$ $\qquad$ $\vec{V} = \sqrt{a^2+b^2}$

iv. Trignometric form

$\vec{V} = V\cos\theta + i\, b\sin\theta = \boxed{V[\cos\theta + i\sin\theta]}$

v. Exponential form

$\vec{V} = V e^{i\theta}$ where $e^{i\theta} = \cos\theta + i\sin\theta$

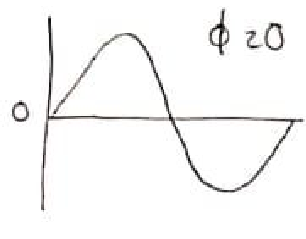
Phase Difference

The phase of an alternating quantity at any instant is the angle ϕ traveled by that alternating qty upto the instant of consideration measured from referance.

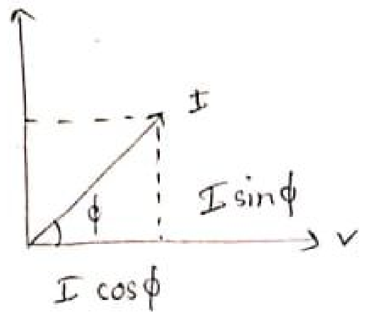
Con 3 cases :

→ case : 1

When $\phi = 0$



Power in AC Circuit



Current I leads voltage by an angle ϕ, by resolving I, $I\cos\phi$ component along V or in phase with V is known as inphase component. $I\sin\phi$ component is in quadrature with V is known as quadrature component

i. Active power / Real power / Useful power

This is the actual power dissipated in circuit resistance. It is given by product of voltage and inphase component of current.

$$\boxed{P = VI\cos\phi}$$ unit = watt

ii. Reactive power

This is the power developed in inductive or capacitive reactance with the circuit. It is given by product of voltage and $\perp r$ component of current.

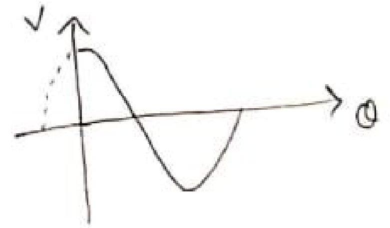
$$\boxed{Q = VI\sin\phi}$$ unit → Volt ampere reactive [VAR]

iii. Apparent power

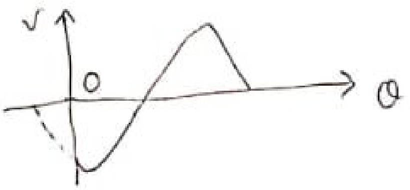
It is the pdt of rms value of voltage and current

$$\boxed{\text{App power} = V_{rms} I_{rms}}$$ unit = VA

case 2: when ϕ is +ve

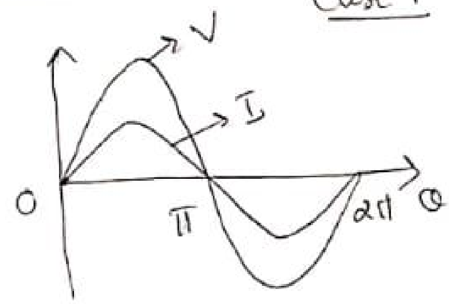


case 3: when ϕ is -ve



Phasor

Case 1

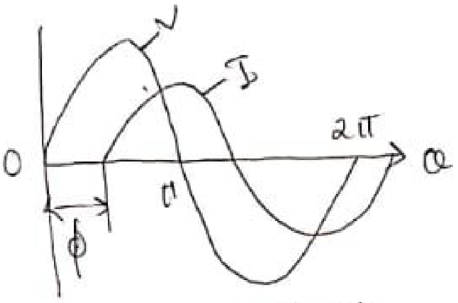


$V_1 = V_m \sin\omega t$

$I = I_m \sin\omega t$

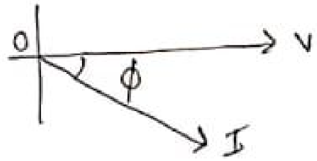
$\boxed{I_m = V_m/R}$

Case 2



$V = V_m \sin\omega t$

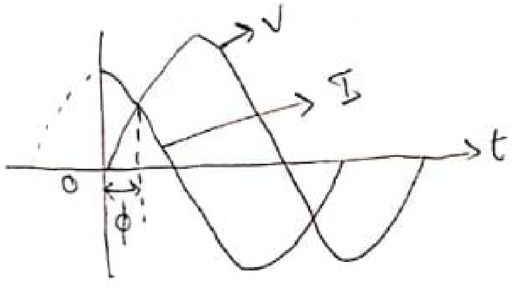
$I = I_m \sin(\omega t - \phi)$



I lags behind V

V leads I

Case 3

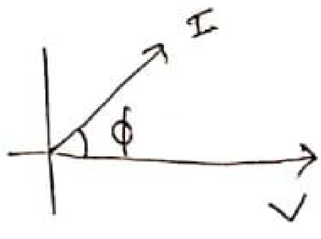


$V = V_m \sin\omega t$

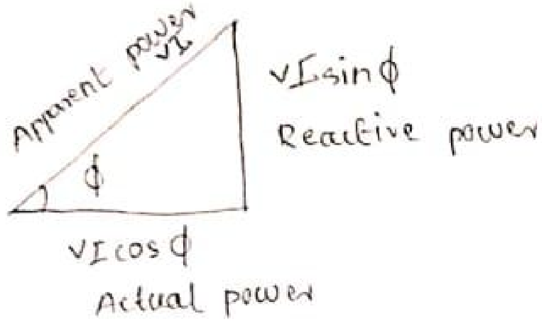
$I = I_m \sin(\omega t + \phi)$

I leads V

V lags I



Power Triangle



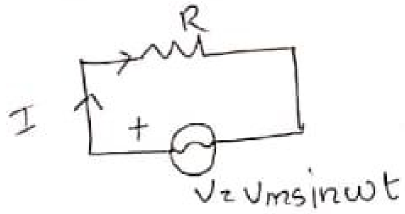
$$\text{Apparent power} = \sqrt{\left(\begin{matrix}\text{Actual}\\ \text{power}\end{matrix}\right)^2 + \left(\begin{matrix}\text{Reactive}\\ \text{power}\end{matrix}\right)^2}$$

$$= \sqrt{(VI\cos\phi)^2 + (VI\sin\phi)^2} = \sqrt{(VI)^2\,[\cos^2\phi + \sin^2\phi]}$$

$$= \sqrt{(VI^2)\,[\cos^2\phi + \sin^2\phi]} = \sqrt{V^2 I^2 \times 1} = VI$$

Q. AC circuit containg resistance only

con an ac circeit consist of pure resistane only connected to an alternating voltage $V = V_m \sin\omega t$

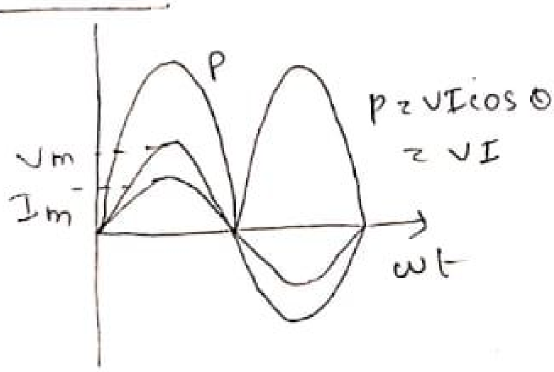


Instantaneous value of current is given by $I = V/R = \frac{V_m \sin\omega t}{R}$

$$= \frac{V_m}{R}\sin\omega t = I_m \sin\omega t$$

where $\left[I_m = \frac{V_m}{R}\right]$

Waveform



Phasor

does not contain any resistive element, hence entire applied voltage has to overcome back emf alone.

i.e $V_m \sin\omega t = L\frac{di}{dt}$ ← Back emf

$L\,di = V_m \sin\omega t \, dt \qquad di = \frac{V_m}{L} \sin\omega t \, dt$

Integrating,

$\int di = \int \frac{V_m}{L} \sin\omega t \, dt \qquad I = \frac{V_m}{L} \times -\frac{\cos\omega t}{\omega}$

$= \frac{V_m}{\omega L} (-\cos\omega t) = V_m/\omega L \sin(\omega t - \pi/2)$

$\sin(A-B) = \sin A \cos B - \cos A \sin B$

$\sin(\omega t - \pi/2) = \sin\omega t \cos\pi/2 - \cos\omega t \sin\pi/2 = 0 - \cos\omega t$

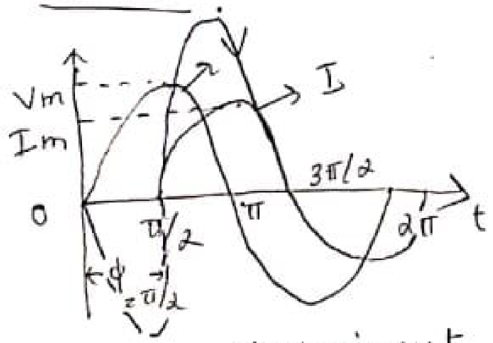
$\therefore I = V_m/\omega L \sin(\omega t - \pi/2)$

$I = \frac{V_m}{\omega L} \sin(\omega t - \pi/2) \qquad I = I_m \sin(\omega t - \pi/2)$

where $I_m = V_m/\omega L = V_m/X_L$

where $\boxed{X_L = \omega L = 2\pi f L}$ → Reactance of inductor

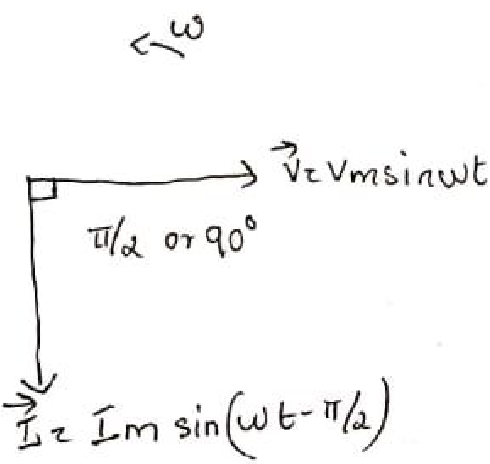
Waveform



$V = V_m \sin\omega t$

$I = I_m \sin(\omega t - \pi/2)$

Phasor



Power

power is drawn from the circuit at any instant is the product of instantaneous voltage and instantaneous current.

$$P = V \times I$$

$$= V_m \sin\omega t \, I_m \sin\omega t = V_m I_m \sin^2\omega t$$

$$= V_m I_m \left(\frac{1-\cos 2\omega t}{2}\right) = \frac{V_m I_m}{2} - \frac{V_m I_m}{2}\cos 2\omega t$$

$$= \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} - \frac{V_m I_m}{2}\cos 2\omega t$$

$$= V_{rms} \times I_{rms} - \frac{V_m I_m}{2}\cos 2\omega t$$

Total power consist of two part

i. constant power i.e $V_{rms} \times I_{rms}$

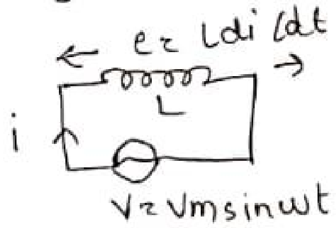
ii. fluctuating power i.e $\frac{V_m I_m}{2}\cos 2\omega t$

The fluctuating power varies at a frequency $\cos 2\omega t$, over a complete cycle the value of $\frac{V_m I_m}{2}\cos 2\omega t$ is zero hence

$$\boxed{\text{power} = V_{rms} \times I_{rms}}$$

AC circuit containing inductance only

An sinusoidal wave $V = V_m \sin\omega t$ applied to pure inductor of L Henry



Then a back emf is produced due to self Inductance of inductance this back emf at any time t oppose change of current through inductor. The inductor

From the equation of current it is clear that current lags behind voltage by 90°. Quantity $\omega \times L$ or X_L is known as inductive reactance of inductor.

Power in purely inductive circuit $[\sin 2A = 2\sin A \cos A]$

instantaneous power is given by $P = V \times I$

$= V_m \sin\omega t \; I_m \sin(\omega t - \pi/2)$

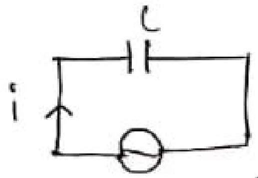
$= V_m I_m \sin\omega t \cdot \sin(\omega t - \pi/2) = V_m I_m \sin\omega t(-\cos\omega t)$

$= V_m I_m \dfrac{\sin(2\omega t)}{2}$

Average power for one complete cycle is zero because avg value of $\sin 2\omega t$ is zero.

∴ Total power consumed by a purely inductive circuit is zero.

AC Circuit containing Capacitance only



$i = C\, dv/dt$

$V = V_m \sin\omega t$

The alternating current passed through the ~~curr~~ circuit is equal to $i = \dfrac{C\,dv}{dt}$

$= C \, d/dt \, (V_m \sin\omega t)$

$= C \; V_m \cos\omega t \times \omega$

$= V_m \omega C \cos\omega t = V_m \omega C \sin(\omega t + \pi/2)$

$= \dfrac{V_m}{1/\omega C} \sin(\omega t + \pi/2) = \dfrac{V_m}{X_C} \sin(\omega t + \pi/2)$

$\vec{V}_R = \vec{I} R$ and $\vec{V}_L = i\vec{I} X_L$

$$\vec{V} = \sqrt{\vec{V}_R^2 + V_L^2} = \sqrt{(IR)^2 + (IX_L)^2} = \sqrt{I^2 (R^2 + X_L^2)}$$

$$V = I\sqrt{R^2 + X_L^2} = I\sqrt{Z^2} = IZ \quad \text{where } Z = \sqrt{R^2 + X_L^2}$$

$Z \rightarrow$ impedence.

$$\vec{V} = \vec{V_R} + \vec{V_L}$$
$$= \vec{I} R + i\vec{I} X_L = \vec{I}(R + iX_L)$$
$$= \vec{I}\vec{Z}$$

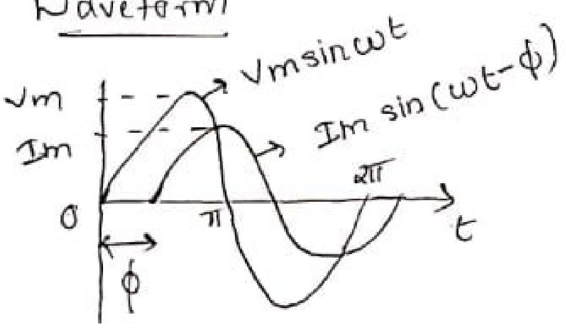
where $\boxed{\vec{Z} = R + iX_L}$ $\quad |\vec{Z}| = \sqrt{R^2 + X_L^2}$

Here $V = V_m \sin\omega t$ and $I = I_m \sin(\omega t - \phi)$

where $\boxed{\phi = \tan^{-1}\left(\frac{X_L}{R}\right)}$ $\quad \tan\phi = V_L/V_R = \frac{IX_L}{IR} = \frac{X_L}{R}$

$\cos\phi = \frac{\vec{V}_R}{\vec{V}} = \frac{\vec{I}R}{\vec{I}Z} = \frac{R}{Z}$ $\quad \boxed{\cos\phi = \frac{R}{Z}}$ is known as power factor of RL circuit

Waveform



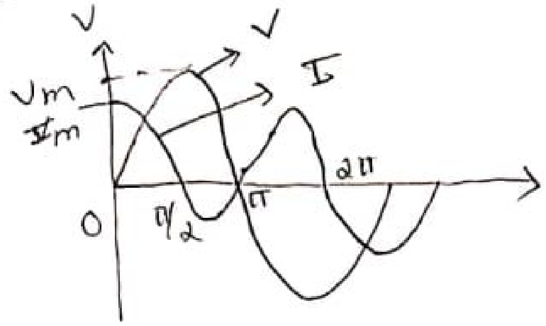
Power in RL circuit

$$\text{power} = VI = I_m\sin(\omega t - \phi)\, V_m \sin\omega t$$
$$= I_m V_m \sin\omega t \sin(\omega t - \phi)$$
$$= I_m V_m \left[\frac{\cos\phi - \cos(2\omega t - \phi)}{2}\right] \quad \because \left(2\sin A\cos B = \cos(A-B) - \cos(A+B)\right)$$
$$= \frac{V_m I_m \cos\phi}{2} - \frac{V_m I_m \cos(2\omega t - \phi)}{2}$$

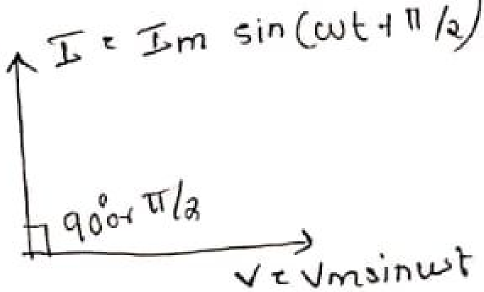
where $I_m = \frac{V_m}{1/\omega C} = V_m/X_C$ where $X_C = 1/\omega C \rightarrow$ Capacitive reactance.

$$\boxed{I = I_m \sin(\omega t + \pi/2)}$$

Wave form



Phasor



Power

$p = V \times I = V_m \sin\omega t \times I_m \sin(\omega t + \pi/2)$

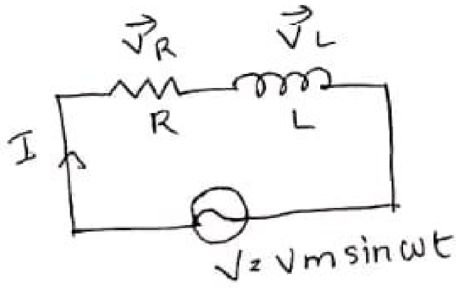
$= V_m I_m \sin\omega t \sin(\omega t + \pi/2)$

$= . 2\, V_m I_m \sin\omega t\,(+\cos\omega t)$

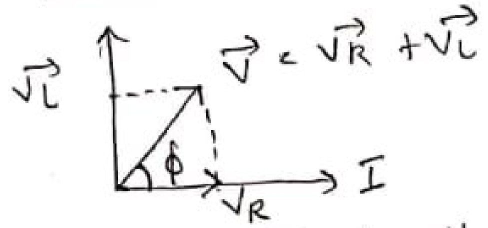
$= p = V_m I_m \frac{\sin 2\omega t}{2}$

Avg power for one cycle - is zero since avg value of sin 2ωt is zero. i.e total power consumed by a purely capacitive circuit e is zero

AC Circuit Containg Resistance and Indutance

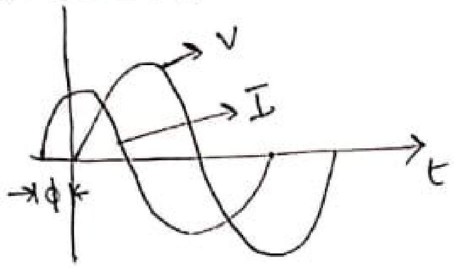


Phasor



Voltage applied is $V = V_m \sin\omega t$. let I be the current flowing through the circuit. Voltage across R is V_R and voltage across L is V_L.

Wave form



Power

$P = V I = V_m \sin\omega t \times I_m \sin(\omega t + \phi)$

$P = I_m V_m \sin\omega t \sin(\omega t + \phi) = V_m I_m \left[\frac{\cos\phi - \cos(2\omega t + \phi)}{2} \right]$

$\left[\because 2\sin A \sin B = \cos(A-B) - \cos(A+B) \right]$

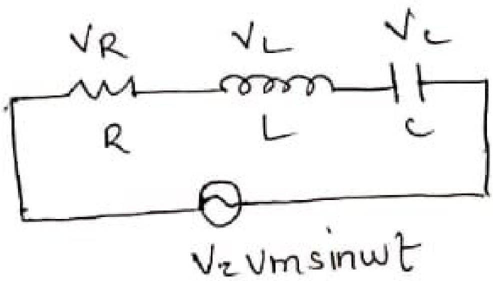
$= \frac{V_m I_m}{2} \cos\phi - \frac{V_m I_m \cos(2\omega t + \phi)}{2}$

Power consist of two part constant power $\frac{V_m I_m \cos\phi}{2}$ and double freq component $\frac{V_m I_m \cos(2\omega t + \phi)}{2}$ whose avg value over a complete cycle is zero.

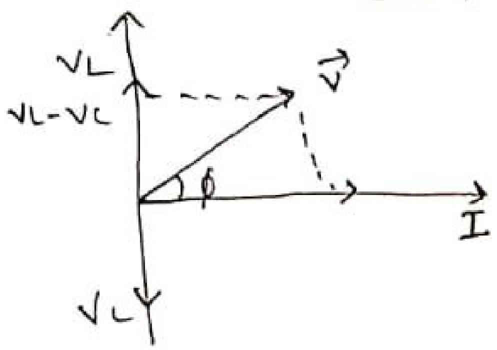
$\therefore$ power consumed in an RC circuit equal to $\frac{V_m I_m \cos\phi}{2}$

$= I_{rms} V_{rms} \cos\phi$

AC circuit containing RLC



Phasor. $X_L > X_C$



Instantaneous power consist of two part

i- Constant power $\frac{V_m I_m \cos\phi}{2}$

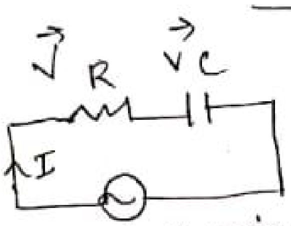
ii- A double frequency component $\frac{V_m I_m \cos(2\omega t - \phi)}{2}$ whose avg value over a complete cycle is zero.

$$\therefore \quad P = \frac{V_m I_m \cos\phi}{2} = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos\phi$$

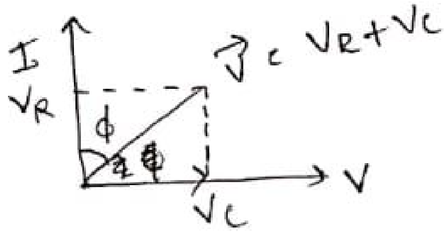
$$= \boxed{V_{rms} I_{rms} \cos\phi}$$

i.e the power consumed by RL circuit

AC Circuit Containing Resistance and Capacitance



$V = V_m \sin\omega t$



$V_R = \vec{I} R \qquad \vec{V}_C = -i \vec{I} X_C$

$$V = \sqrt{V_R^2 + V_C^2} \left[= \sqrt{IR^2 + (IX_C)^2} = \sqrt{I^2 (R^2 + X_C^2)} \right.$$

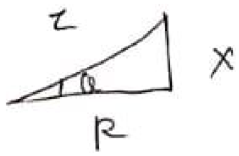
$$V = I \sqrt{R^2 + X_C^2} \Big]$$

$$\vec{V} = \vec{I} R + -i\vec{I} X_C = \vec{I} [R - iX_C] = \vec{I} \vec{Z}$$

$$\boxed{\vec{Z} = R - iX_C}$$

$$Z^2 = \sqrt{R^2 + X_C^2} \qquad \cos\phi = R/Z \qquad \phi = \tan^{-1} X_C/R$$

Impedance see:



Here $V = V_m \sin\omega t \qquad I = I_m \sin(\omega t + \phi)$

$12.12 \angle 22.83\ \Omega$

Power factor $= \cos\phi$

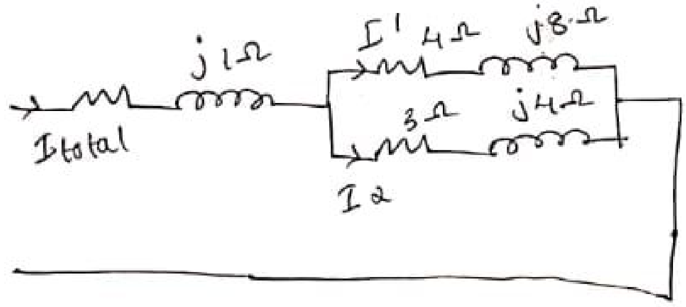
$= \cos(22.8) = 0.921$ //

Power consumed $= VI\cos\phi$

$= (50\angle 36.8 \times 4.12 \angle 14.03) \times 0.92$

$= \underline{118.47 + 145.45i}$

Q.



Find I_{total}, I_1, I_2, Z_{total}, power consumed by circuit

ans

$Z_{tot} = Z_1 + Z_2 // Z_3$

$= (2+j1) + \dfrac{(4+j8)(3+j4)}{(4+j8+3+j4)} = \dfrac{726}{193} + \dfrac{713}{193}i$

$= 5.27\angle 44.48$ //

$I_{total} = V/Z_{tot} = \dfrac{200}{5.27\angle 44.48} = \underline{26.92 - 26.74i}$

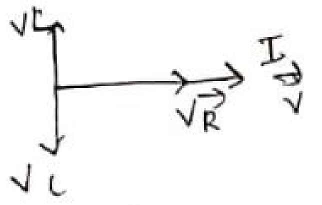
$= 37.94 \angle -44.48$

$I_1 = \dfrac{(26.92 - 26.74i) \times (3+j4)}{(3+j4) + (4+j8)} = \underline{8.51 - 10.67i}$

$I_2 = \dfrac{(26.92 - 26.74i) \times (4+j8)}{(4+j8+3+j4)}$ $= 13.64 \angle -51.42$ //

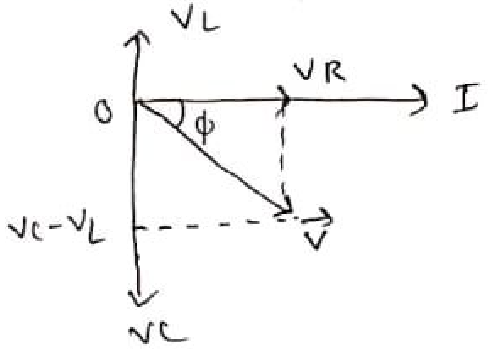
$= 18.40 - 16.06i = 24.42\angle -41.1$ //

$X_L = X_C$



$\phi = 0$

$X_C > X_L$



Resonance

case I : $\vec{V} = \vec{V_R} + \vec{V_L} - \vec{V_C}$

case II : $\vec{V} = \vec{V_R}$

case III : $\vec{V_R} + (\vec{V_C} - \vec{V_L})$ $\quad \vec{V} = I(R \pm jx)$

$I\,[R \pm i(X_L - X_C)] = IZ$

Q. The current flowing through a circuit $I(t) = 25\sin(\omega t - 30)$ $v(t) = 100\sin\omega t$ when a voltage of $v(t)$ is applied find active and reactive power.

ans. Active power $= VI\cos\phi$

reactive power $= VI\sin\phi$

$$AP = \frac{25}{\sqrt{2}} \times \frac{\sqrt{3}}{2} \times \frac{100}{\sqrt{2}} = \frac{25 \times \cancel{100}^{50} \times \sqrt{3}}{2 \times \cancel{2}} = 625\sqrt{3}$$

$$RP = \frac{25}{\sqrt{2}} \times \frac{100}{\sqrt{2}} \times \frac{1}{2} = \frac{25 \times \cancel{100}^{50}}{2 \times \cancel{2}} = 625$$

Q A 50 Hz sinusoidal voltage $(40 + 30j)$ is applied to a series RL circuit resulting in sinusoidal current $(4 + 1j)$ calculate impedance of circuit, power factor and power consumed

ans $V = 40 + 30j$ $\quad V = IZ$

$I = 4 + 1j$ $\quad Z = V/I = \frac{40 + 30j}{4 + 1j} = \frac{190}{17} + \frac{80}{17}j$

$= 11.17 + 4.7i$

Power consumed $= 200 \times 37.94 \times \cos(44.45)$

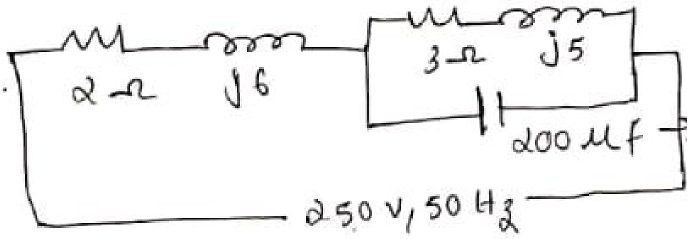
$= 5414.00$

$Z_{eff} \rightarrow$ polar form $\angle \theta \leftarrow \phi$

$\downarrow$

mag Z_{eff} $\quad I_{tot} = V/Z_{eff}$

Q. Find tot I, power factor and power consumed



$\rightarrow -15.92j$

$X_C = 1/\omega C$

$= \frac{1}{2\pi \times 50 \times 200 \times 10^{-6}}$

$= 15.92$

$$Z_{tot} = 2 + j6 + \frac{(3+j5) \times (-15.92j)}{(3+j5 - 15.92j)}$$

$= 7.92 + 11.66i = 14.10 \angle 55.785$

$$I_{total} = \frac{250}{14.10} = 17.75\,A$$

Power factor $= \cos\phi = \cos(55.785) = 0.5622$

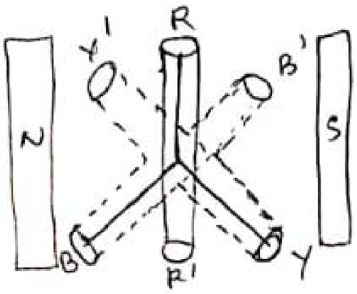
power consumed $= VI\cos\phi$

$= 250 \times 17.75 \times 0.56$

$= 2492.39\ W$

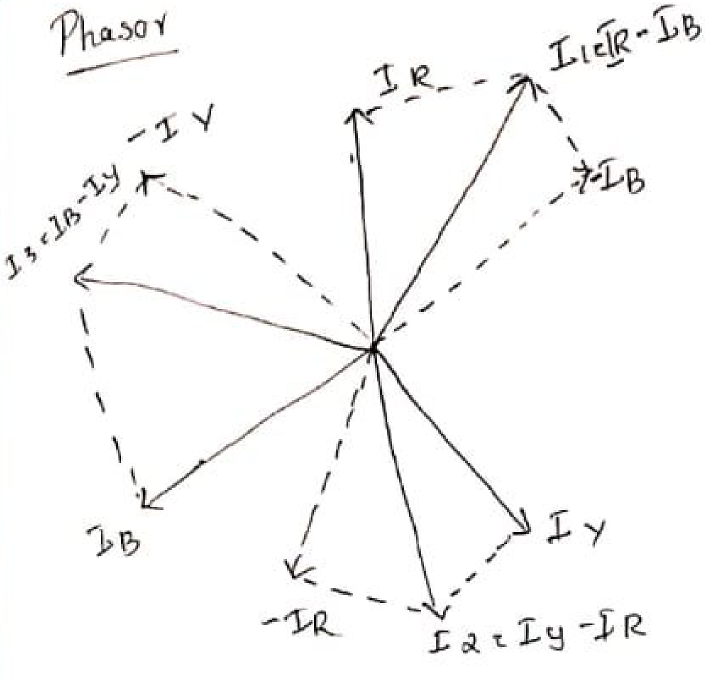
Threee phase System

Generation of 3 phase AC voltage



The fig show an elementary 3 phase alternator, R, R', Y, Y' are placed at 120° from each other and rotated in a uniform mag field, sinusoidal voltage is induced in each coil will be of same magnitude and frequency

Phasor



$I_1 = I_R - I_B$ $\quad I_1 = \sqrt{I_R^2 + I_B^2 + 2 I_R I_B \cos 60}$

$I_L = \sqrt{I_{ph}^2 + I_{ph}^2 + 2 I_{ph} I_{ph} \cos 60} = \sqrt{2 I_{ph}^2 + 2 I_p^2 \times 1/2}$

$= \sqrt{3 I_{ph}^2} = \sqrt{3} I_{ph}$ //

Relation b/w phase and line voltages

$V_{RY} = V_{RN}$ $\quad V_{YB} = V_{YN}$ $\quad V_{BR} = V_{BN}$

i.e $\boxed{V_L = V_{ph}}$

Power

tot power = 3 × power in each phase

$P = 3 \times V_{ph} I_{ph} \cos\phi$

$= 3 \times V_L \times \frac{I_L}{\sqrt{3}} \cos\phi = \sqrt{3} V_L I_L \cos\phi$

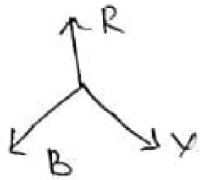
Q. A star connected 3 phase load has a resistance of 12 Ω and inductive reactance of 16 Ω in each branch a line to line voltage of 415 volt is applied find phase voltage, phase current, line current power factor and total power.

Equation for 3 induced voltages are $V_R = V_m \sin \omega t$

$V_Y = V_m \sin(\omega t - 120^\circ)$ $V_B = V_m \sin(\omega t - 240^\circ)$

Phase Sequence

The phase sequence of 3 phase system give the order in which voltage in 3 phases reach their max value.



RYB

In this system two letters are used as suffix of symbol for voltage or current. The order of writing subscript indicate the direction in which voltage act or current flows

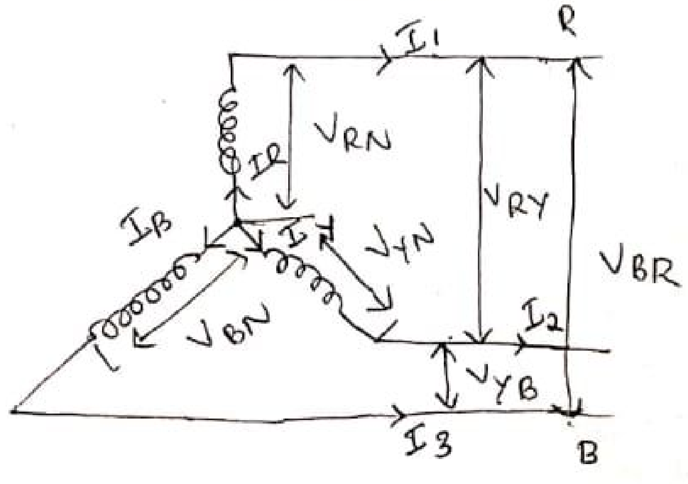
The representation V_{RY} means that voltage from point R to Y i.e R is a higher potential compared to Y and $V_{RY} = V_{RN} + V_{NY}$ and $V_{YB} = V_{YN} - V_{BN}$

$V_{BR} = V_{BN} - V_{RN}$

Methods of Connection of 3 phase system

There are two methods:

i. Star Connection or Y connection



Relation b/w line and phase current

$I_1 = I_R \quad I_2 = I_Y \quad I_3 = I_B$

i.e $\boxed{I_L = I_{ph}}$

Power

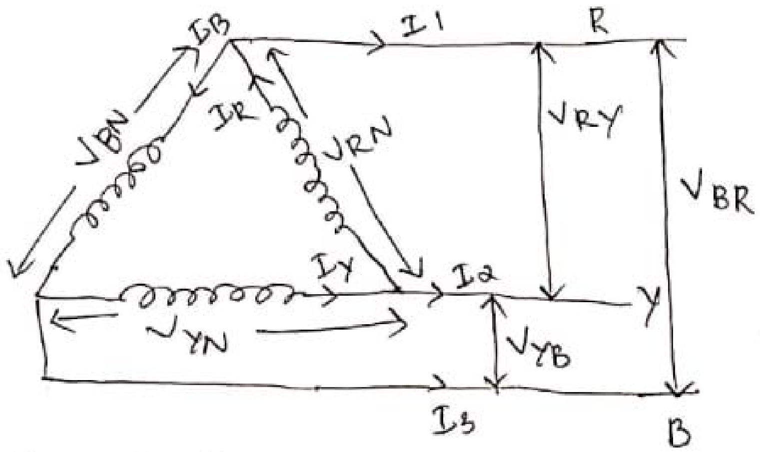
tot Power = 3× power in each phase

$$\boxed{P = 3 \times V_{ph} I_{ph} \cos\phi}$$

$$P = 3 \times \frac{V_L}{\sqrt{3}} I_L \cos\phi = \boxed{\sqrt{3}\, V_L I_L \cos\phi = P}$$

Delta Connection

In delta connection dissimiliar ends of 3 phase winding are joined together i.e finishing end of one phase is connected to starting end of other phase



$I_R = I_1 + I_B$

$\boxed{I_1 = I_R - I_B}$ → Relation b/w line and phase current

$I_2 + I_R = I_Y$ (my) Phasor

$\boxed{I_2 = I_Y - I_R}$

$I_3 + I_Y = I_B$

$\boxed{I_3 = I_B - I_Y}$

In this method similiar ends of 3 phases are connected together to form a common junction called neutral point (N). The voltage between any line and neutral point i.e voltage across each winding is known as phase voltage. The voltage b/w any two line is known as line voltages

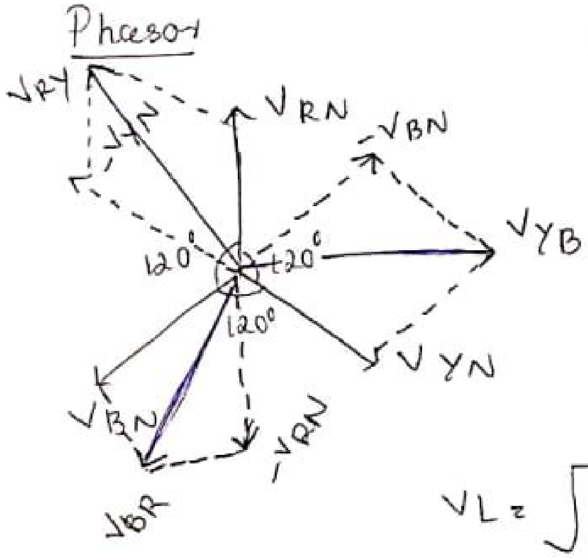
∴ Phase voltages are: V_{RN}, V_{YN}, V_{BN}

Line voltages are: V_{RY}, V_{YB}, V_{BR}

when the load across the system is balanced, the neutral wire carries no current i.e $I_N = 0$

Relation b/w Line and phase voltage

Phasor



$$V_{RY} = V_{RN} - V_{YN}$$

$$V_{YB} = V_{YN} - V_{BN}$$

$$V_{BR} = V_{BN} - V_{RN}$$

$$V_{RY} = \sqrt{(V_{RN})^2 + (V_{YN})^2 + 2V_{RN}V_{YN}\cos 60}$$

$$V_L = \sqrt{V_{ph}^2 + V_{ph}^2 + 2V_{ph}V_{ph}\cos 60}$$

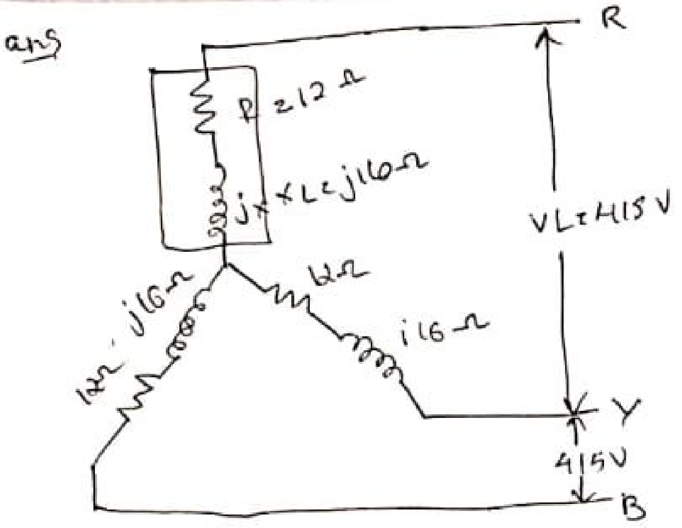
$$= \sqrt{2V_{ph}^2 + 2V_{ph}^2 \times 1/2} = \sqrt{2V_{ph}^2 + V_{ph}^2}$$

$$= V_L = \sqrt{3V_{ph}^2} \quad \boxed{V_L = \sqrt{3}\,V_{ph}}$$

∴ For a star connection line voltage equal to $\sqrt{3}$ phase voltage. All line voltages are equal in magnitude for a balanced system.

For a start

ans



$V_L = \sqrt{3}\, V_{ph}$

$I_L = I_{ph}$

$P = 3\, V_{ph} I_{ph} \cos\phi$

$= \sqrt{3}\, V_L I_L \cos\phi$

i. phase voltage, $V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{415}{\sqrt{3}} = 239.60$

ii. Phase current, $I_{ph} = \frac{V_{ph}}{Z} = \frac{239.6}{(12+j16)} = 251.6 + 16i$

$= 252.10 < 3.63$

iii. Line current, $I_L = I_{ph} = 251.6 + 16i$

iv. Power $= 3\, V_{ph} I_{ph} \cos\phi = 3 \times 239.6 \times 11.9 \times 0.6 =$

v. $\cos\phi = R/Z = \frac{12}{\sqrt{12^2+16^2}} = 0.6$

Q. 3 equal impedence of $(4+3j)\Omega$ are connected in delta across 200V supply find line current, power factor and tot power